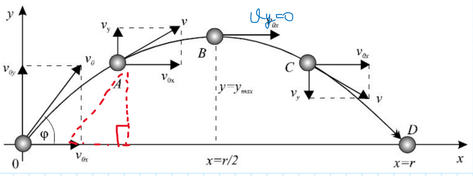
Ligjerata 3dh

Hedhja e pjerrët



- horizontale
- vertikale

$\cos\varphi = \frac{v_{0x}}{v_0}$

$v_{0x} = v_x = v_0\cos\varphi$ → komponenta e shpejtësisë në drejtimin të x

$\sin\varphi = \frac{v_{0y}}{v_0} \Rightarrow v_{0y} = v_0\sin\varphi$

$v_y = v_{0y} - gt = v_0\sin\varphi - gt$ → komponenta e shpejtësisë në drejtim të y

$v_x = v_0\cos\varphi$
$v_y = v_0\sin\varphi - gt$

$v_x = \frac{dx}{dt} \rightarrow \frac{dx}{dt} = v_0\cos\varphi \,/\, dt$

$dx = v_0\cos\varphi\, dt \,/\, \int$

$\int dx = v_0\cos\varphi \int dt$

$x = v_0 t\cos\varphi$

$v_y = \frac{dy}{dt} \rightarrow \frac{dy}{dt} = v_0\sin\varphi - gt \,/\, dt$

$dy = v_0\sin\varphi\, dt - gt\, dt \,/\, \int$

$\int dy = v_0\sin\varphi \int dt - g\int t\, dt$

$y = v_0 t\sin\varphi - \frac{gt^2}{2}$

$x = v_0 t\cos\varphi \rightarrow t = \frac{x}{v_0\cos\varphi}$

$y = v_0 t\sin\varphi - \frac{gt^2}{2}$

$y = \frac{v_0}{v_0}\frac{x}{\cos\varphi}\cdot\sin\varphi - \frac{g}{2}\cdot\frac{x^2}{v_0^2\cos^2\varphi}$

$$y = x\cdot tg\varphi - \frac{g}{2v_0^2\cos^2\varphi}\cdot x^2$$

ekuacioni i hedhjes së pjerrët.

$x = x_{max}$
$y = 0$

$0 = x_{max}\, tg\varphi - \frac{g}{2v_0^2\cos^2\varphi}x_{max}^2$

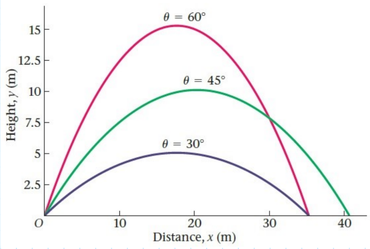
$x_{max}\, tg\varphi = \frac{g}{2v_0^2\cos^2\varphi}x_{max}^2$

$tg\varphi = \frac{g}{2v_0^2\cos^2\varphi}\cdot x_{max} \,/\cdot \frac{2v_0^2\cos^2\varphi}{g}$ $\quad tg\varphi = \frac{\sin\varphi}{\cos\varphi}$

$x_{max} = \frac{2v_0^2\cos^2\varphi\cdot\sin\varphi}{g\cdot\cos\varphi} = \frac{2v_0^2\sin\varphi\cos\varphi}{g}$; $2\sin\varphi\cos\varphi = \sin 2\varphi$

$$x_{max} = \frac{v_0^2\sin 2\varphi}{g}$$

Largësia maksimale.



$v_0 = 10\frac{m}{s}$ $\quad \varphi = 30°,\ \varphi = 60°$

I) $x_{max} = \frac{100\frac{m^2}{s^2}\cdot\sin 60°}{9{,}81\frac{m}{s^2}} = 8{,}82m$

II) $x_{max} = \frac{100\frac{m^2}{s^2}\cdot\sin 120°}{9{,}81\frac{m}{s^2}} = 8{,}82m$

D.sh.
$\varphi = 10°,\ \varphi = 80°$
$\varphi = 20°,\ \varphi = 70°$

$v_0 = 10\frac{m}{s}$ $\quad \varphi = 45°$ $\quad x_{max} = \frac{100\frac{m^2}{s^2}\cdot\sin 90°}{9{,}81\frac{m}{s^2}} = \frac{100}{9{,}81}m = 10{,}19m$

$y = y_{max}$; $x = \frac{x_{max}}{2}$

$y_{max} = \frac{x_{max}}{2}tg\varphi - \frac{g}{2v_0^2\cos^2\varphi}\cdot\frac{x_{max}^2}{4}$

$y_{max} = \frac{v_0^2\sin 2\varphi}{2g}\cdot\frac{\sin\varphi}{\cos\varphi} - \frac{g}{2v_0^2\cos^2\varphi}\cdot\frac{v_0^4\sin^2 2\varphi}{4g^2}$

$y_{max} = \frac{2v_0^2\sin\varphi\cos\varphi}{2g\cos\varphi} - \frac{v_0^2\sin^2 2\varphi\cdot g}{8v_0^2\cos^2\varphi\cdot g}$

$y_{max} = \frac{v_0^2\sin 2\varphi}{g} - \frac{v_0^2\sin 2\varphi}{2g}$

$$y_{max} = \frac{v_0^2\sin 2\varphi}{2g}$$

Lartësia maksimale.

$\varphi = ?$

$x_{max} = y_{max}$ ose $\boxed{r = q}$

Shtim

$\frac{v_0^2\sin 2\varphi}{g} = \frac{v_0^2\sin^2\varphi}{2g}$

$2\sin\varphi\cos\varphi = \frac{\sin\varphi}{2} \,/\cdot\frac{2}{\cos\varphi}$

$2\cos\varphi\cdot\frac{2}{\cos\varphi} = \frac{\sin\varphi}{2}\cdot\frac{2}{\cos\varphi}$

$4 = tg\varphi$

$\varphi = \arctg 4$ $\quad \boxed{\varphi = 75{,}96°}$

$y = v_0 t\sin\varphi - \frac{gt^2}{2}$
$y = 0,\ t = t_{x_{max}}$

$0 = v_0 t_{x_{max}}\sin\varphi - g\frac{t_{x_{max}}^2}{2}$

$v_0 t_{x_{max}}\sin\varphi = g\frac{t_{x_{max}}}{2} \,/\cdot\frac{2}{g}$

$$\frac{2v_0\sin\varphi}{g} = t_{x_{max}}$$

$v_y = v_0\sin\varphi - gt$

$v_y = 0 \quad t = t_{y_{max}}$

$0 = v_0\sin\varphi - g t_{y_{max}}$

$v_0\sin\varphi = g t_{y_{max}} \,/\, g$

$$t_{y_{max}} = \frac{v_0\sin\varphi}{g}$$

$$t_{y_{max}} = \frac{t_{x_{max}}}{2}$$

ose

$$t_{x_{max}} = 2t_{y_{max}}$$